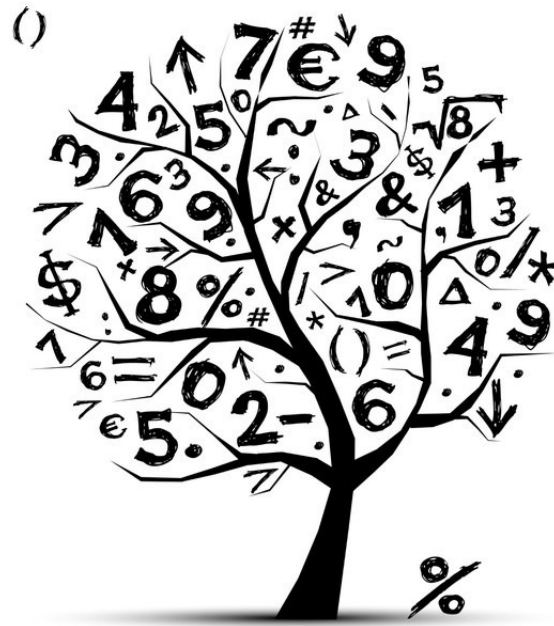


Last update: October 7, 2026

Eigenvalues and eigenvectors



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1 Eigenvalues

Definition 1: eigenvalue and eigenvector

Let $Q \in \mathbb{R}^{n \times n}$ be a square matrix. A scalar $\lambda \in \mathbb{R}$ is called an **eigenvalue** of Q if there exists a nonzero vector $v \in \mathbb{R}^n \setminus \{0\}$ such that

$$Qv = \lambda v.$$

In this case, v is called an **eigenvector** associated with λ .

- Eigenvalues are solutions of the **characteristic equation**

$$\det(Q - \lambda I) = 0.$$

- If Q is symmetric, then all its eigenvalues are real.
- The function $p(\lambda) = \det(Q - \lambda I)$ is a polynomial of degree n in λ , called the **characteristic polynomial** of Q .
- If λ is an eigenvalue, the eigenvectors associated with λ are the nonzero solutions v of the homogeneous system

$$(Q - \lambda I)v = 0.$$
- If v is an eigenvector associated with λ , then also αv is an eigenvector associated with λ , for every $\alpha \in \mathbb{R} \setminus \{0\}$. Eigenvectors are therefore never unique: we just give one of them (“for example”).

A square symmetric matrix $Q \in \mathbb{R}^{n \times n}$ is

- positive semidefinite if and only if all its eigenvalues are non-negative.
- positive definite if and only if all its eigenvalues are strictly positive.
- negative semidefinite if and only if all its eigenvalues are non-positive.
- negative definite if and only if all its eigenvalues are strictly negative.

- Recall that (semi)definiteness has been defined in the chapter on matrices through the sign of $x'Qx$: for instance, Q is positive semidefinite if $x'Qx \geq 0$ for all $x \in \mathbb{R}^n$. The box above gives an equivalent characterization in terms of eigenvalues.

- One direction is easy to see: if $\mathbf{Q}\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$, then

$$\mathbf{v}'\mathbf{Q}\mathbf{v} = \lambda\mathbf{v}'\mathbf{v} = \lambda \sum_{i=1}^n v_i^2, \quad \text{with} \quad \sum_{i=1}^n v_i^2 > 0,$$

so $\mathbf{v}'\mathbf{Q}\mathbf{v}$ and λ have the same sign. For example, if $\mathbf{Q}$ is positive semidefinite, then every eigenvalue satisfies $\lambda \geq 0$.

- A symmetric matrix which is neither positive semidefinite nor negative semidefinite is called **indefinite**: equivalently, it has at least one strictly positive and at least one strictly negative eigenvalue.

Example 1: eigenvalues and eigenvectors

Let us consider the matrix

$$\mathbf{Q} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The characteristic equation is:

$$\det(\mathbf{Q} - \lambda\mathbf{I}) = \det \begin{pmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{pmatrix} = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3.$$

Hence

$$\det(\mathbf{Q} - \lambda\mathbf{I}) = 0 \iff (\lambda - 1)(\lambda - 3) = 0,$$

and therefore the eigenvalues are:

$$\lambda_1 = 1, \quad \lambda_2 = 3.$$

An eigenvector associated with $\lambda_1 = 1$ can be found by solving

$$(\mathbf{Q} - \mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0},$$

which gives (for example)

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Similarly, an eigenvector associated with $\lambda_2 = 3$ satisfies

$$(\mathbf{Q} - 3\mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \mathbf{v} = \mathbf{0},$$

which gives (for example)

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Therefore,

$$\mathbf{Q}\mathbf{v}_1 = 1\mathbf{v}_1, \quad \mathbf{Q}\mathbf{v}_2 = 3\mathbf{v}_2.$$

We can verify it directly:

$$\mathbf{Q}\mathbf{v}_1 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1\mathbf{v}_1, \quad \mathbf{Q}\mathbf{v}_2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix} = 3\mathbf{v}_2.$$

Since $\mathbf{Q}$ is symmetric and both eigenvalues are strictly positive, the matrix $\mathbf{Q}$ is **positive definite**.

2 Properties of eigenvalues

2.1 Triangular and diagonal matrices

Observation 1: eigenvalues of triangular and diagonal matrices

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be an upper triangular, a lower triangular or a diagonal matrix. Then the eigenvalues of $\mathbf{Q}$ are its diagonal entries $q_{11}, q_{22}, \dots, q_{nn}$.

Proof. If $\mathbf{Q}$ is triangular (or diagonal), then also $\mathbf{Q} - \lambda \mathbf{I}$ is triangular (or diagonal), with diagonal entries $q_{11} - \lambda, \dots, q_{nn} - \lambda$. Recall that the determinant of a triangular matrix is the product of its diagonal entries. Hence

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = \prod_{i=1}^n (q_{ii} - \lambda),$$

which is equal to zero if and only if $\lambda = q_{ii}$ for some $i \in \{1, 2, \dots, n\}$. □

Example 2: eigenvalues of a triangular matrix

Let us consider the upper triangular matrix

$$\mathbf{U} = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 4 \\ 0 & 0 & 5 \end{pmatrix}.$$

The characteristic equation is

$$\det(\mathbf{U} - \lambda \mathbf{I}) = \det \begin{pmatrix} 2 - \lambda & 1 & 3 \\ 0 & -1 - \lambda & 4 \\ 0 & 0 & 5 - \lambda \end{pmatrix} = (2 - \lambda)(-1 - \lambda)(5 - \lambda) = 0,$$

and therefore the eigenvalues are the diagonal entries:

$$\lambda_1 = 2, \quad \lambda_2 = -1, \quad \lambda_3 = 5.$$

2.2 Trace, determinant and eigenvalues

Definition 2: trace of a matrix

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be a square matrix. The **trace** of $\mathbf{Q}$ is the sum of its diagonal entries:

$$\text{tr}(\mathbf{Q}) = \sum_{i=1}^n q_{ii} = q_{11} + q_{22} + \dots + q_{nn}.$$

- The characteristic polynomial $p(\lambda) = \det(\mathbf{Q} - \lambda \mathbf{I})$ has degree n . Over the complex numbers, a polynomial of degree n has exactly n roots $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{C}$, provided that each root is counted as many times as its **multiplicity** (the number of times the factor $(\lambda_i - \lambda)$ appears in the factorization). Hence

$$p(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \cdots (\lambda_n - \lambda).$$

- The real roots are the eigenvalues of Definition 1. The roots which are not real are called **complex eigenvalues** of $\mathbf{Q}$ (their eigenvectors have complex entries). If $\mathbf{Q}$ is symmetric, all the roots are real.

Observation 2: determinant, trace and eigenvalues

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and let $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{C}$ be the roots of its characteristic polynomial, each repeated according to its multiplicity. Then

$$\det(\mathbf{Q}) = \prod_{i=1}^n \lambda_i, \quad \text{tr}(\mathbf{Q}) = \sum_{i=1}^n \lambda_i.$$

Idea of the proof. Setting $\lambda = 0$ in $p(\lambda) = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$ gives $\det(\mathbf{Q}) = p(0) = \lambda_1 \lambda_2 \cdots \lambda_n$. For the trace, we compare the coefficients of λ^{n-1} : in the product $(\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$ this coefficient is $(-1)^{n-1} \sum_{i=1}^n \lambda_i$, while in $\det(\mathbf{Q} - \lambda \mathbf{I})$ the power λ^{n-1} only comes from the product of the diagonal entries $(q_{11} - \lambda) \cdots (q_{nn} - \lambda)$, and its coefficient is $(-1)^{n-1} \sum_{i=1}^n q_{ii}$. $\square$

Example 3: trace and determinant

- For the matrix of Example 1 we have $\lambda_1 = 1$ and $\lambda_2 = 3$, and indeed

$$\text{tr} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 2 + 2 = 4 = 1 + 3, \quad \det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 4 - 1 = 3 = 1 \cdot 3.$$

- Multiplicities must be taken into account. The diagonal matrix

$$\mathbf{D} = \begin{pmatrix} 4 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

has characteristic polynomial $(4 - \lambda)^2(-3 - \lambda)$: the eigenvalue 4 has multiplicity 2 and the eigenvalue -3 has multiplicity 1. Hence $\lambda_1 = \lambda_2 = 4$, $\lambda_3 = -3$, and

$$\text{tr}(\mathbf{D}) = 4 + 4 - 3 = 5, \quad \det(\mathbf{D}) = 4 \cdot 4 \cdot (-3) = -48.$$

- Complex roots must be taken into account. For the (non symmetric) matrix

$$\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{we have} \quad \det(\mathbf{R} - \lambda \mathbf{I}) = \lambda^2 + 1,$$

which has no real roots: $\mathbf{R}$ has no (real) eigenvalues. Its complex eigenvalues are $\lambda_1 = i$ and $\lambda_2 = -i$, and indeed

$$\text{tr}(\mathbf{R}) = 0 = i + (-i), \quad \det(\mathbf{R}) = 1 = i \cdot (-i).$$

2.3 Eigenvectors of distinct eigenvalues

- Recall that the vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent if $\alpha_1 \mathbf{v}_1 + \dots + \alpha_k \mathbf{v}_k = \mathbf{0}$ implies $\alpha_1 = \dots = \alpha_k = 0$.

Observation 3: eigenvectors of distinct eigenvalues

Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be **distinct** eigenvalues of $\mathbf{Q} \in \mathbb{R}^{n \times n}$ and let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ be associated eigenvectors ($\mathbf{Q}\mathbf{v}_i = \lambda_i \mathbf{v}_i$). Then $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ are linearly independent.

- In particular, if $\mathbf{Q} \in \mathbb{R}^{n \times n}$ has n distinct real eigenvalues, then one eigenvector for each eigenvalue gives n linearly independent vectors of $\mathbb{R}^n$.
- In Example 1, the eigenvectors $\mathbf{v}_1 = (1, 1)'$ and $\mathbf{v}_2 = (1, -1)'$ are linearly independent, since neither of them is a multiple of the other.

3 A complete example with a 3×3 matrix

Example 4: eigenvalues and eigenvectors of a 3×3 matrix

Let us consider the symmetric matrix

$$\mathbf{Q} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}.$$

Characteristic polynomial. Expanding along the first row (Laplace expansion):

$$\begin{aligned} \det(\mathbf{Q} - \lambda \mathbf{I}) &= \det \begin{pmatrix} 2-\lambda & 1 & 0 \\ 1 & 3-\lambda & 1 \\ 0 & 1 & 2-\lambda \end{pmatrix} \\ &= (2-\lambda)[(3-\lambda)(2-\lambda) - 1] - 1 \cdot [1 \cdot (2-\lambda) - 1 \cdot 0] \\ &= (2-\lambda)[(3-\lambda)(2-\lambda) - 2] = (2-\lambda)(\lambda^2 - 5\lambda + 4) \\ &= (2-\lambda)(\lambda-1)(\lambda-4). \end{aligned}$$

Therefore the eigenvalues are

$$\lambda_1 = 1, \quad \lambda_2 = 2, \quad \lambda_3 = 4.$$

Eigenvector for $\lambda_1 = 1$. We solve

$$(\mathbf{Q} - \mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff \begin{cases} v_1 + v_2 = 0 \\ v_1 + 2v_2 + v_3 = 0 \\ v_2 + v_3 = 0 \end{cases}$$

From the first and the third equation, $v_1 = -v_2$ and $v_3 = -v_2$ (the second equation is then satisfied). Choosing $v_2 = -1$:

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}.$$

Eigenvector for $\lambda_2 = 2$. We solve

$$(\mathbf{Q} - 2\mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff \begin{cases} v_2 = 0 \\ v_1 + v_2 + v_3 = 0 \end{cases}$$

hence $v_2 = 0$, $v_3 = -v_1$ and, for example,

$$\mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

Eigenvector for $\lambda_3 = 4$. We solve

$$(\mathbf{Q} - 4\mathbf{I})\mathbf{v} = \mathbf{0} \iff \begin{pmatrix} -2 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & 1 & -2 \end{pmatrix} \mathbf{v} = \mathbf{0} \iff \begin{cases} -2v_1 + v_2 = 0 \\ v_1 - v_2 + v_3 = 0 \\ v_2 - 2v_3 = 0 \end{cases}$$

From the first and the third equation, $v_2 = 2v_1$ and $v_3 = v_2/2 = v_1$ (the second equation is then satisfied: $v_1 - 2v_1 + v_1 = 0$). Choosing $v_1 = 1$:

$$\mathbf{v}_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}.$$

Checks. By Observation 2,

$$\text{tr}(\mathbf{Q}) = 2 + 3 + 2 = 7 = 1 + 2 + 4, \quad \det(\mathbf{Q}) = 2 \cdot (6 - 1) - 1 \cdot (2 - 0) = 8 = 1 \cdot 2 \cdot 4.$$

Since the three eigenvalues are distinct, by Observation 3 the eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly independent. Finally, $\mathbf{Q}$ is symmetric with strictly positive eigenvalues, hence it is **positive definite**.

4 Sylvester's criterion

- Computing all the eigenvalues of a matrix can be difficult. For symmetric matrices, definiteness and semidefiniteness can also be checked by computing determinants of suitable square submatrices.

Definition 3: principal minors

Let $Q \in \mathbb{R}^{n \times n}$.

- A **principal minor** of order k of Q is the determinant of the $k \times k$ submatrix obtained from Q by keeping the rows and the columns with the **same** indices $i_1 < i_2 < \dots < i_k$ (equivalently, by deleting the same $n - k$ rows and columns).
- The **leading principal minor** of order k is the principal minor obtained by keeping the first k rows and columns:

$$\Delta_k = \det \begin{pmatrix} q_{11} & \cdots & q_{1k} \\ \vdots & \ddots & \vdots \\ q_{k1} & \cdots & q_{kk} \end{pmatrix}, \quad k = 1, 2, \dots, n.$$

- A matrix $Q \in \mathbb{R}^{n \times n}$ has exactly n leading principal minors $\Delta_1 = q_{11}, \Delta_2, \dots, \Delta_n = \det(Q)$.
- For $n = 3$, the principal minors are: of order 1, q_{11}, q_{22}, q_{33} ; of order 2, the determinants of the submatrices with indices $\{1, 2\}, \{1, 3\}, \{2, 3\}$; of order 3, $\det(Q)$.

Theorem 1: Sylvester's criterion

Let $Q \in \mathbb{R}^{n \times n}$ be a **symmetric** matrix with leading principal minors $\Delta_1, \Delta_2, \dots, \Delta_n$. Then

- Q is **positive definite** if and only if

$$\Delta_k > 0, \quad \forall k \in \{1, 2, \dots, n\};$$

- Q is **negative definite** if and only if the leading principal minors alternate in sign starting with a negative one:

$$\Delta_1 < 0, \quad \Delta_2 > 0, \quad \Delta_3 < 0, \quad \dots \quad \text{i.e.} \quad (-1)^k \Delta_k > 0, \quad \forall k \in \{1, 2, \dots, n\}.$$

Observation 4: semidefinite matrices and principal minors

Let $Q \in \mathbb{R}^{n \times n}$ be a symmetric matrix. Then

- Q is **positive semidefinite** if and only if **all** its principal minors (not only the leading ones)

are ≥ 0 ;

- $\mathbf{Q}$ is **negative semidefinite** if and only if every principal minor of order k has the sign of $(-1)^k$ or is zero, i.e., $(-1)^k$ (principal minor of order k) ≥ 0 , for all k .

For semidefiniteness it is **not** enough to check $\Delta_k \geq 0$ for the leading principal minors only.

Example 5: a positive definite matrix

For the matrix of Example 4

$$\mathbf{Q} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

the leading principal minors are

$$\Delta_1 = 2 > 0, \quad \Delta_2 = \det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 6 - 1 = 5 > 0, \quad \Delta_3 = \det(\mathbf{Q}) = 8 > 0.$$

By Sylvester's criterion, $\mathbf{Q}$ is positive definite, in agreement with its eigenvalues 1, 2, 4.

Example 6: an indefinite matrix

Let us consider

$$\mathbf{Q} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

We have $\Delta_1 = 1 > 0$ and $\Delta_2 = 1 - 4 = -3 < 0$. Hence:

- $\mathbf{Q}$ is not positive definite ($\Delta_2 < 0$) and not negative definite ($\Delta_1 > 0$);
- $\mathbf{Q}$ is not positive semidefinite (the principal minor Δ_2 of order 2 is negative) and not negative semidefinite ($(-1)^2 \Delta_2 < 0$).

Therefore $\mathbf{Q}$ is **indefinite**. Indeed, recalling the definition via $\mathbf{x}'\mathbf{Q}\mathbf{x}$:

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}: \mathbf{x}'\mathbf{Q}\mathbf{x} = 1 > 0, \quad \mathbf{x} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}: \mathbf{x}'\mathbf{Q}\mathbf{x} = 1 - 4 + 1 = -2 < 0.$$

The eigenvalues are the roots of $(1 - \lambda)^2 - 4 = (\lambda - 3)(\lambda + 1)$, i.e., 3 and -1 : one positive and one negative.

Example 7: a singular positive semidefinite matrix

Let us consider

$$\mathbf{Q} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

The leading principal minors are $\Delta_1 = 1$, $\Delta_2 = 2 - 1 = 1$ and $\Delta_3 = \det(\mathbf{Q}) = 1 \cdot (2 - 1) - 1 \cdot (1 - 0) = 0$. Since $\Delta_3 = 0$, $\mathbf{Q}$ is **not** positive definite.

To decide whether it is positive semidefinite we check **all** principal minors:

- order 1: $q_{11} = 1$, $q_{22} = 2$, $q_{33} = 1$;
- order 2: $\det \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} = 1$, $\det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1$, $\det \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = 1$;
- order 3: $\det(\mathbf{Q}) = 0$.

All principal minors are ≥ 0 , hence $\mathbf{Q}$ is **positive semidefinite**. Indeed

$$\mathbf{x}'\mathbf{Q}\mathbf{x} = x_1^2 + 2x_1x_2 + 2x_2^2 + 2x_2x_3 + x_3^2 = (x_1 + x_2)^2 + (x_2 + x_3)^2 \geq 0,$$

and the eigenvalues are 0, 1, 3 ($\mathbf{v} = (1, -1, 1)'$ satisfies $\mathbf{Q}\mathbf{v} = \mathbf{0} = 0\mathbf{v}$).

Example 8: leading principal minors are not enough for semidefiniteness

Let us consider

$$\mathbf{Q} = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}.$$

The leading principal minors are $\Delta_1 = 0 \geq 0$ and $\Delta_2 = 0 \geq 0$. However $\mathbf{Q}$ is **not** positive semidefinite: for

$$\mathbf{x} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{we have} \quad \mathbf{x}'\mathbf{Q}\mathbf{x} = -1 < 0.$$

The principal minor $q_{22} = -1$ (which is not a leading one) detects this. In fact, the eigenvalues are 0 and -1 , and $\mathbf{Q}$ is negative semidefinite.